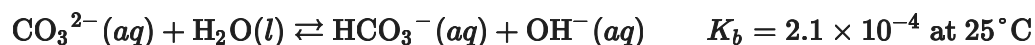


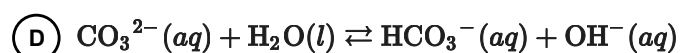
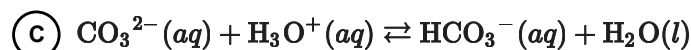
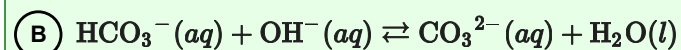
Unit 8 Progress Check: MCQ

Buffer solutions containing Na_2CO_3 and NaHCO_3 range in **pH** from 10.0 to 11.0. The chemical equation below represents the equilibrium between CO_3^{2-} and H_2O , and the table lists the composition of four different buffer solutions at 25°C .



Buffer	$[\text{NaHCO}_3]$	$[\text{Na}_2\text{CO}_3]$	pH
1	0.150	0.100	?
2	0.200	0.200	10.32
3	0.100	0.100	10.32
4	0.100	0.200	?

1. Which of the following chemical equilibrium equations best shows what happens in the buffer solutions to minimize the change in **pH** when a small amount of a strong base is added?



2. Which of the following mathematical expressions can be used to determine the approximate **pH** of buffer 1?



Unit 8 Progress Check: MCQ

(A) $\text{pH} = -\log(2.1 \times 10^{-4}) + \log\left(\frac{0.100}{0.150}\right) = 3.50$

(B) $\text{pH} = -\log(2.1 \times 10^{-4}) + \log\left(\frac{0.150}{0.100}\right) = 3.85$

(C) $\text{pH} = [14.00 + \log(2.1 \times 10^{-4})] + \log\left(\frac{0.100}{0.150}\right) = 10.15$ ✓

(D) $\text{pH} = [14.00 + \log(2.1 \times 10^{-4})] + \log\left(\frac{0.150}{0.100}\right) = 10.50$

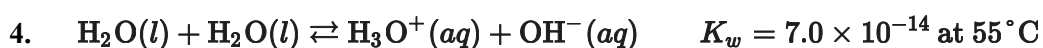
3. Which mathematical expression can be used to explain why buffer 2 and buffer 3 have the same pH?

(A) $\log\left(\frac{0.200}{0.200}\right) = \log\left(\frac{0.100}{0.100}\right) = \log(1)$ ✓

(B) $-\log(K_w) - [-\log(K_b)] = \text{p}K_a$

(C) $0.200\text{ M} = 2 \times (0.100\text{ M})$

(D) $0.200\text{ M} - 0.200\text{ M} = 0.100\text{ M} - 0.100\text{ M}$



Pure water autoionizes as shown in the equation above. Based on this information, which of the following is correct?

(A) The autoionization equilibrium for pure water favors the formation of reactants at 55°C compared to 25°C .

(B) The autoionization equilibrium for pure water produces the same amount of OH^- ions at 55°C and 25°C .

(C) At 55°C , $\text{pH} = -\log(\sqrt{K_w})$ for pure water. ✓

(D) At 55°C , $\text{pH} = -\log(K_w)$ for pure water.



Unit 8 Progress Check: MCQ

5. $K_w = [\text{H}_3\text{O}^+][\text{OH}^-] = 1.0 \times 10^{-14}$ at 25°C

Based on the information above, which of the following is true for a sample of pure water at 25°C ?

- (A) $[\text{H}_3\text{O}^+] = 7.0\text{ M}$
- (B) $[\text{OH}^-] = 1.0 \times 10^{-14}\text{ M}$
- (C) $\text{pH} = 10^{-7}$

- (D) $\text{pOH} = 7.00$



6. $2\text{H}_2\text{O}(l) \rightleftharpoons \text{H}_3\text{O}^+(aq) + \text{OH}^-(aq) \Delta H^\circ = +56\text{ kJ/mol}_{\text{rxn}}$

The endothermic autoionization of pure water is represented by the chemical equation shown above. The pH of pure water is measured to be 7.00 at 25.0°C and 6.02 at 100.0°C . Which of the following statements best explains these observations?

- (A) At the higher temperature water dissociates less, $[\text{H}_3\text{O}^+] < [\text{OH}^-]$, and the water becomes basic.
- (B) At the higher temperature water dissociates less, $[\text{H}_3\text{O}^+] = [\text{OH}^-]$, and the water remains neutral.
- (C) At the higher temperature water dissociates more, $[\text{H}_3\text{O}^+] > [\text{OH}^-]$, and the water becomes acidic.

- (D) At the higher temperature water dissociates more, $[\text{H}_3\text{O}^+] = [\text{OH}^-]$, and the water remains neutral.



7. Which of the following gives the best estimate for the pH of a $5 \times 10^{-4}\text{ M Sr(OH)}_2(aq)$ solution at 25°C ?



Unit 8 Progress Check: MCQ

(A) $\text{pH} \approx 3.0$ because $\text{Sr}(\text{OH})_2$ is a strong acid.

(B) $\text{pH} \approx 5.0$ because $\text{Sr}(\text{OH})_2$ is a weak acid.

(C) $\text{pH} \approx 9.0$ because $\text{Sr}(\text{OH})_2$ is a weak base.

(D) $\text{pH} \approx 11.0$ because $\text{Sr}(\text{OH})_2$ is a strong base. ✓

8. Which of the following gives the best estimate for the pH of a $1 \times 10^{-5} \text{ M HClO}_4(\text{aq})$ solution at 25°C ?

(A) $\text{pH} \approx 1.0$ because HClO_4 is a strong acid.

(B) $\text{pH} \approx 5.0$ because HClO_4 is a strong acid. ✓

(C) $\text{pH} \approx 7.0$ because ClO_4^- is a strong base.

(D) $\text{pH} \approx 9.0$ because ClO_4^- is a strong base.

9. Which of the following is the correct mathematical relationship to use to calculate the pH of a 0.10 M aqueous HBr solution?

(A) $\text{pH} = [\text{H}_3\text{O}^+] = 0.10$

(B) $\text{pH} = -\log(1.0 \times 10^{-1})$ ✓

(C) $\text{pH} = 7.00 - (0.10)$

(D) $\text{pH} = \log(1.0 \times 10^{-1})$



Unit 8 Progress Check: MCQ

10. $\text{HCOOH}(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{H}_3\text{O}^+(aq) + \text{HCOO}^-(aq)$ $K_a = 1.8 \times 10^{-4}$ at 25°C

Initial $[\text{HCOOH}]$	Percent Ionization	Equilibrium $[\text{H}_3\text{O}^+]$
0.150	3.4%	5.1×10^{-3}
0.100	4.2%	4.2×10^{-3}
0.0500	5.8%	2.9×10^{-3}

The equilibrium for the acid ionization of **HCOOH** is represented by the equation above and the table gives the percent ionization for **HCOOH** at different initial concentrations of the weak acid at 25°C . Based on the information, which of the following is true for a **0.125 M** aqueous solution of **HCOOH**?

- (A) It has a larger percent ionization, a lower $[\text{H}_3\text{O}^+]_{eq}$, and a lower **pH** than a **0.150 M HCOOH** solution does.
- (B) It has a larger percent ionization, a lower $[\text{H}_3\text{O}^+]_{eq}$, and a higher **pOH** than a **0.150 M HCOOH** solution does.
- (C) It has a lower percent ionization, a larger $[\text{H}_3\text{O}^+]_{eq}$, and a higher **pOH** than a **0.100 M HCOOH** solution does. ✓
- (D) It has a lower percent ionization, a larger $[\text{H}_3\text{O}^+]_{eq}$, and a higher **pH** than a **0.100 M HCOOH** solution.

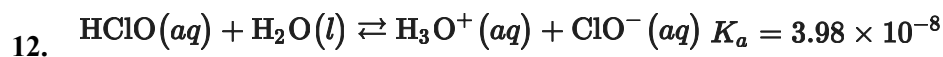
11. $\text{NH}_3(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{NH}_4^+(aq) + \text{OH}^-(aq)$ $K_b = 1.8 \times 10^{-5}$ at 25°C

The reaction between **NH₃** and water is represented above. A solution that is initially **0.200 M** in **NH₃** has a **pH** of 11.28. Which of the following correctly predicts the **pH** of a solution for which $[\text{NH}_3]_{\text{initial}} = 0.100 \text{ M}$, and why?

- (A) The **pH** will be lower than 11.28 because decreasing the initial concentration of **NH₃** increases the equilibrium concentration of the conjugate acid **NH₄⁺**.
- (B) The **pH** will be lower than 11.28 because the equilibrium concentration of **OH⁻** ions decreases when the initial concentration of the base decreases. ✓
- (C) The **pH** will be higher than 11.28 because decreasing the initial concentration of **NH₃** decreases the equilibrium concentration of the conjugate acid **NH₄⁺**.
- (D) The **pH** will be higher than 11.28 because the equilibrium concentration of **OH⁻** ions increases when the initial concentration of the base decreases.



Unit 8 Progress Check: MCQ



The acid ionization equilibrium for HClO at 25°C is shown above. A 0.100 M aqueous solution of this acid has a pH of about 4.2. If a solution with an initial concentration of HClO of 0.200 M is allowed to reach equilibrium at the same temperature, which of the following correctly predicts its pH , and why?

- (A) The pH will be higher than 4.2 because increasing the concentration of the weak acid decreases the percent ionization.
- (B) The pH will be higher than 4.2 because, for weak acid solutions, the pH is directly proportional to the initial concentration of the weak acid.
- (C) The pH will be lower than 4.2 because increasing the concentration of the weak acid produces more H_3O^+ to establish equilibrium. ✓
- (D) The pH will be lower than 4.2 because the K_a of the weak acid is inversely proportional to the initial concentration of the weak acid.

13. The weak acid CH_3COOH has a $\text{p}K_a$ of 4.76. A solution is prepared by mixing $500.\text{ mL}$ of $0.150\text{ M CH}_3\text{COOH}(aq)$ and 0.0200 mol of $\text{NaOH}(s)$. Which of the following can be used to calculate the pH of the solution?

- (A) $\text{pH} = 4.76 + \log\left(\frac{0.0200}{0.150}\right) = 3.88$
- (B) $\text{pH} = 4.76 + \log\left(\frac{0.0200}{0.0750}\right) = 4.19$
- (C) $\text{pH} = 4.76 + \log\left(\frac{0.0200}{0.0550}\right) = 4.32$ ✓
- (D) $\text{pH} = 4.76 + \log\left(\frac{0.0950}{0.0750}\right) = 4.86$

14. Which of the following provides the correct mathematical expression to calculate the pH of a solution made by mixing 10.0 mL of 1.00 M HCl and 11.0 mL of 1.00 M NaOH at 25°C ?



Unit 8 Progress Check: MCQ

(A) $\text{pH} = -\log(0.0010) = 3.00$

(B) $\text{pH} = 14.00 + \log(0.001) = 11.00$

(C) $\text{pH} = 14.00 + \log\left(\frac{0.0010}{0.0210}\right) = 12.68$



(D) $\text{pH} = 14.00 + \log\left(\frac{0.0010}{0.0110}\right) = 12.96$

15.

Substance	$\text{p}K_a$
$\text{NH}_4^+(aq)$	9.25
$\text{HCl}(aq)$	< 1

A buffer solution is formed by mixing equal volumes of $0.12\text{ M NH}_3(aq)$ and $0.10\text{ M HCl}(aq)$, which reduces the concentration of both solutions by one half. Based on the $\text{p}K_a$ data given in the table, which of the following gives the pH of the buffer solution?

(A) $\text{pH} = -\log(0.050) = 1.30$

(B) $\text{pH} = 9.25 + \log\left(\frac{0.010}{0.050}\right) = 8.55$



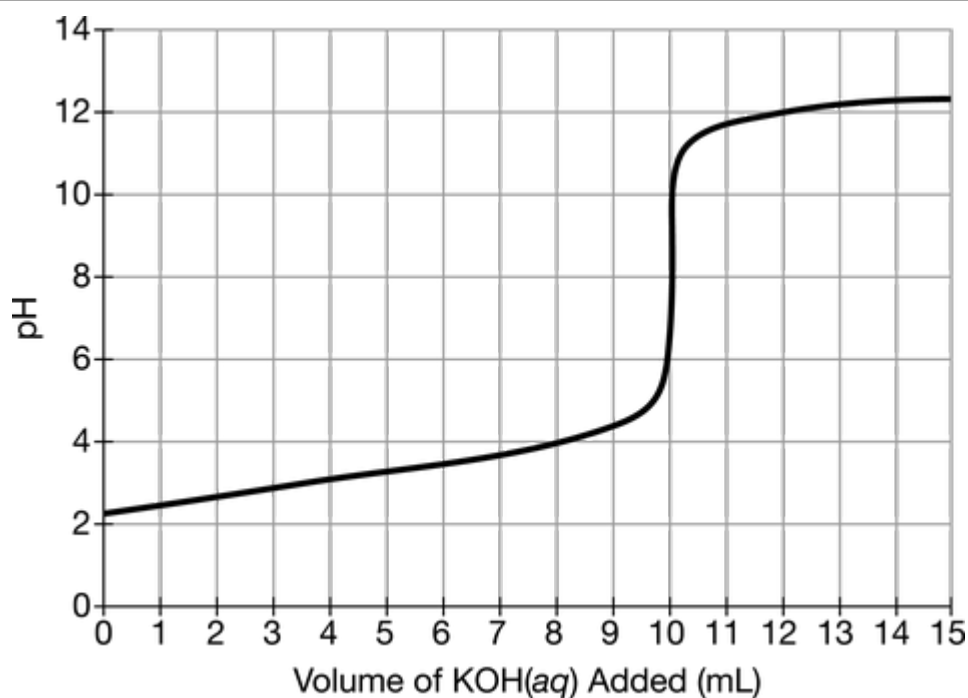
(C) $\text{pH} = 9.25 + \log\left(\frac{0.060}{0.050}\right) = 9.32$

(D) $\text{pH} = 14.00 - (-\log(0.010)) = 12.00$



Unit 8 Progress Check: MCQ

16.



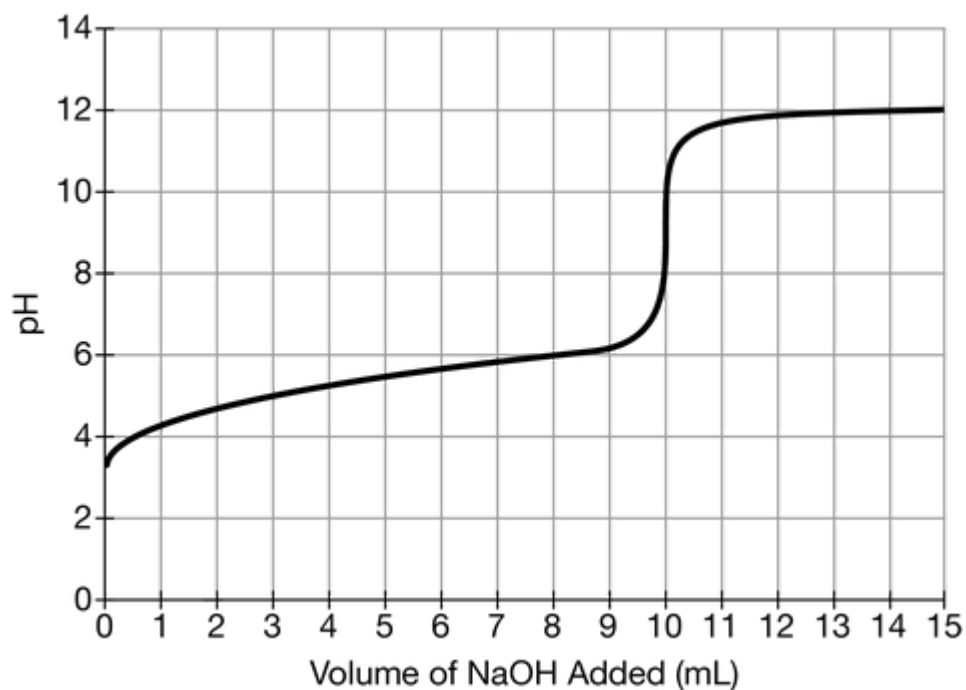
The pH versus volume data for the titration of $0.10\text{ M HNO}_2(\text{aq})$ with $0.10\text{ M KOH}(\text{aq})$ is plotted on the graph above. Based on the data, which of the following species is present in the greatest concentration after 6.0 mL of $\text{KOH}(\text{aq})$ has been added to the solution of $\text{HNO}_2(\text{aq})$?

- (A) $\text{H}^+(\text{aq})$
- (B) $\text{HNO}_2(\text{aq})$
- (C) $\text{NO}_2^-(\text{aq})$
- (D) $\text{OH}^-(\text{aq})$



Unit 8 Progress Check: MCQ

17.



A 0.20 M solution of the weak acid potassium hydrogen phthalate (**KHP**) is titrated with $0.10\text{ M NaOH}(aq)$. Based on the titration curve shown in the graph above, the $\text{p}K_a$ of **KHP** is closest to which of the following?

(A) 3.6

(B) 5.4



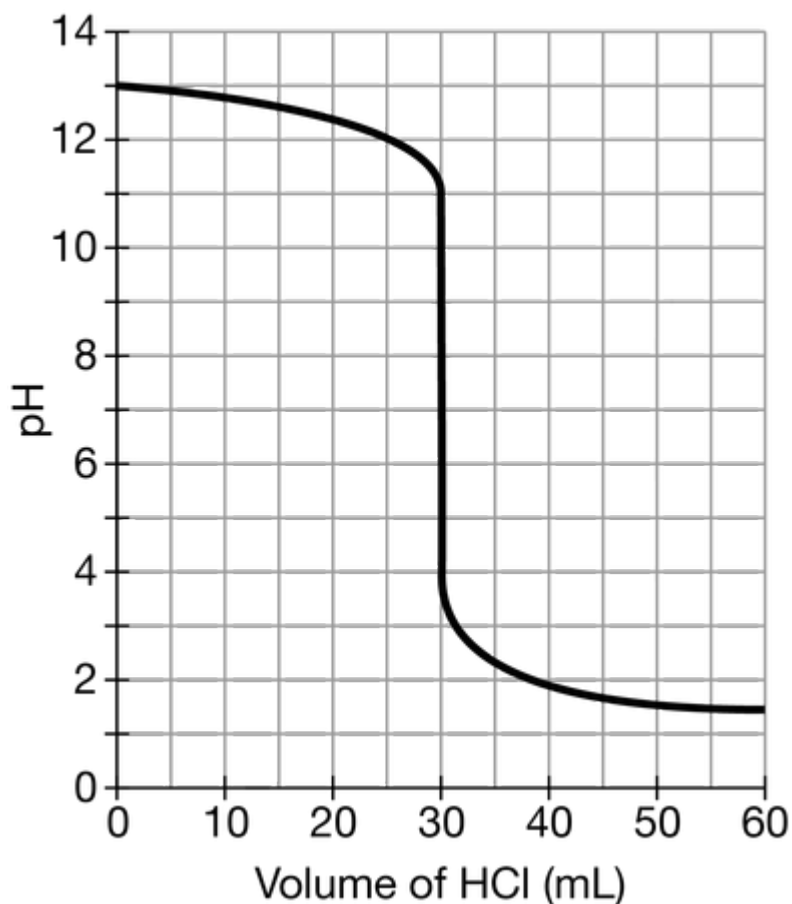
(C) 8.8

(D) 11.8



Unit 8 Progress Check: MCQ

18.



A 60. mL sample of $\text{NaOH}(aq)$ was titrated with $0.10\text{ }M\text{ HCl}(aq)$. Based on the resulting titration curve shown above, what was the approximate concentration of NaOH in the sample?

(A) $0.033\text{ }M$

(B) $0.050\text{ }M$

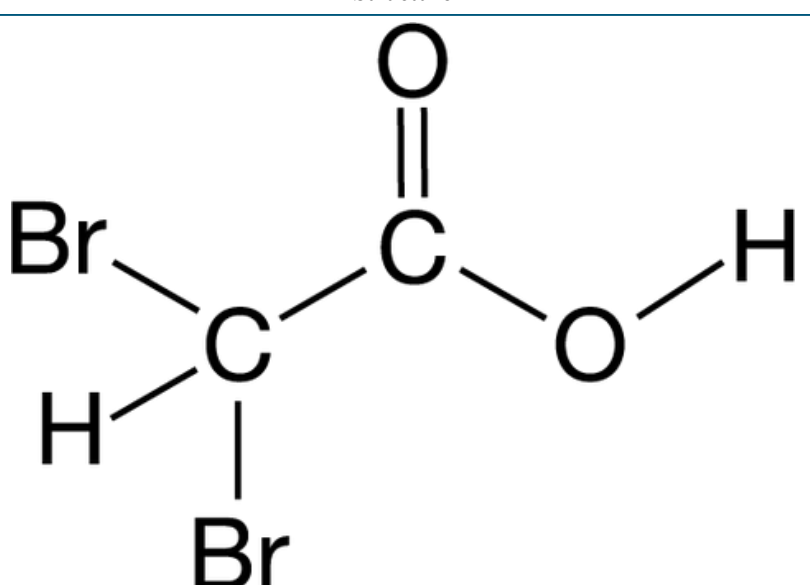
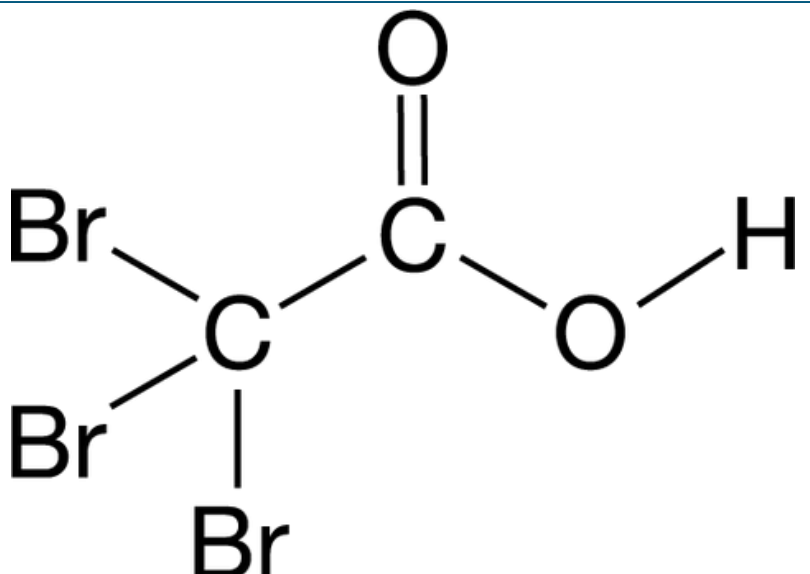


(C) $0.10\text{ }M$

(D) $0.20\text{ }M$



Unit 8 Progress Check: MCQ

19.	Name	Structure	K_a (at 25°C)
	Acid 1		3.3×10^{-2}
	Acid 2		1.9×10^{-1}

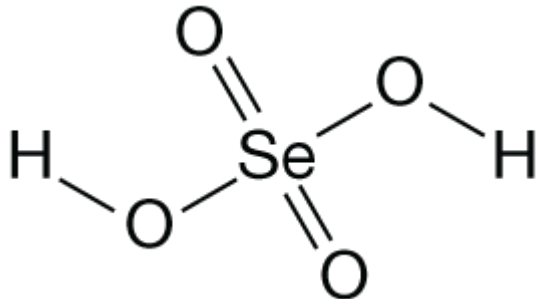
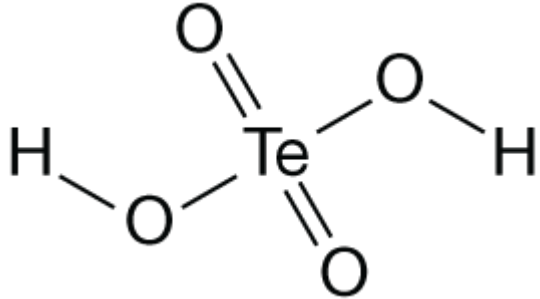
The table above provides information on two weak acids. Which of the following explains the difference in their acid strength?

- (A) Acid 1 is a stronger acid because it has more acidic hydrogen atoms than acid 2.
- (B) Acid 1 is a stronger acid because it can make more hydrogen bonds than acid 2.
- (C) Acid 2 is a stronger acid because it is a larger, more polarizable molecule with stronger intermolecular forces than acid 1.
- (D) Acid 2 is a stronger acid because it has a more stable conjugate base than acid 1 due to the greater number of electronegative Br atoms. ✓



Unit 8 Progress Check: MCQ

20. $\text{H}_2\text{XO}_4(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{H}_3\text{O}^+(aq) + \text{HXO}_4^-(aq) K_{a1}$
 $\text{HXO}_4^-(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{H}_3\text{O}^+(aq) + \text{XO}_4^{2-}(aq) K_{a2}$

Acid	K_{a1}	K_{a2}
	$\gg 1$	2×10^{-2}
	2×10^{-8}	1×10^{-11}

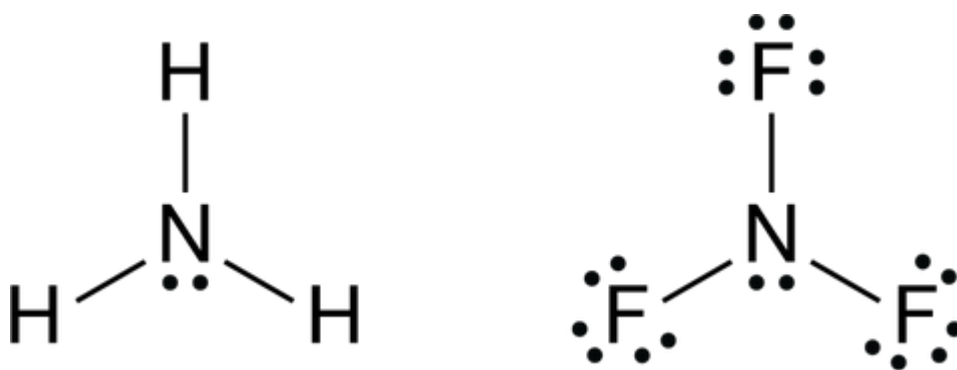
The equilibrium reactions for diprotic oxoacids with a general formula H_2XO_4 are represented by the equations above. The acid ionization constants for H_2SeO_4 and H_2TeO_4 are provided in the table. Which of the following best explains the difference in strength for these two acids?

- (A) H_2SeO_4 is weaker because **Se** has a smaller positive formal charge than **Te**, resulting in a decrease in its ability to transfer an H^+ to H_2O .
- (B) H_2TeO_4 is weaker because **Te** has a smaller positive formal charge than **Se**, resulting in a decrease in its ability to transfer an H^+ to H_2O .
- (C) H_2SeO_4 is weaker because **Se** is more electronegative than **Te**, resulting in more stable conjugate bases HSeO_4^- and SeO_4^{2-} than those for H_2TeO_4 .
- (D) H_2TeO_4 is weaker because **Te** is less electronegative than **Se**, resulting in less stable conjugate bases HTeO_4^- and TeO_4^{2-} than those for H_2SeO_4 . ✓



Unit 8 Progress Check: MCQ

21.



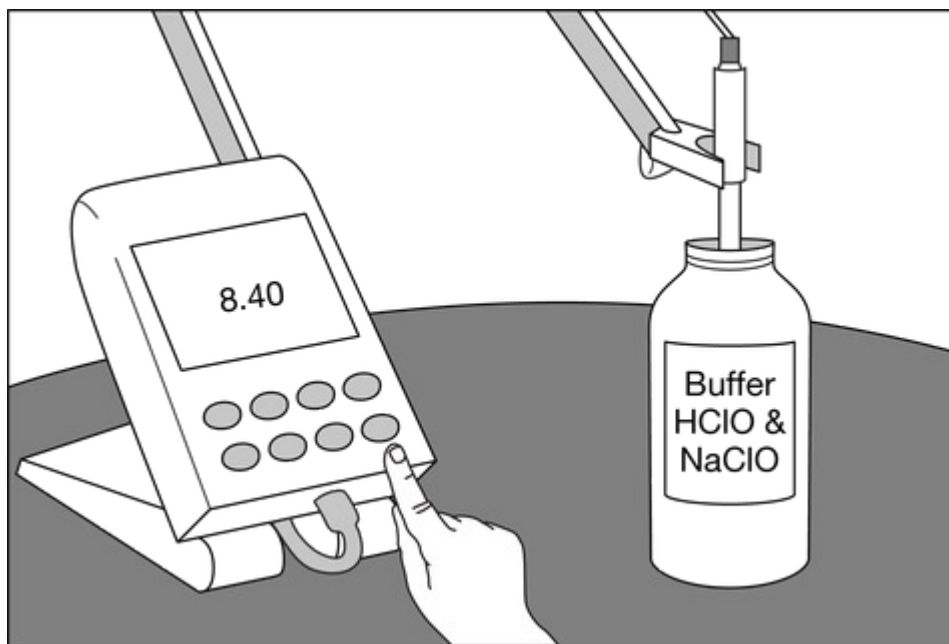
Lewis diagrams of the weak bases NH_3 and NF_3 are shown above. Based on these diagrams, which of the following predictions of their relative base strength is correct, and why?

- (A) NF_3 is a stronger base than NH_3 because more than one resonance structure exists for its conjugate acid NF_3H^+ , making it more stable than NH_4^+ .
- (B) NF_3 is a stronger base than NH_3 because of the greater electronegativity of F compared with H .
- (C) NF_3 is a weaker base than NH_3 because of the greater electronegativity of F compared with H . ✓
- (D) NF_3 is a weaker base than NH_3 because more than one resonance structure exists for its conjugate acid NF_3H^+ , making it more stable than NH_4^+ .



Unit 8 Progress Check: MCQ

22.



A student measures the pH of a 0.0100 M buffer solution made with HClO and NaClO , as shown above. The $\text{p}K_a$ of HClO is 7.40 at 25°C . Based on this information, which of the following best compares the relative concentrations of ClO^- and HClO in the buffer solution?

(A) $[\text{ClO}^-] = [\text{HClO}]$

(B) $[\text{ClO}^-] > [\text{HClO}]$



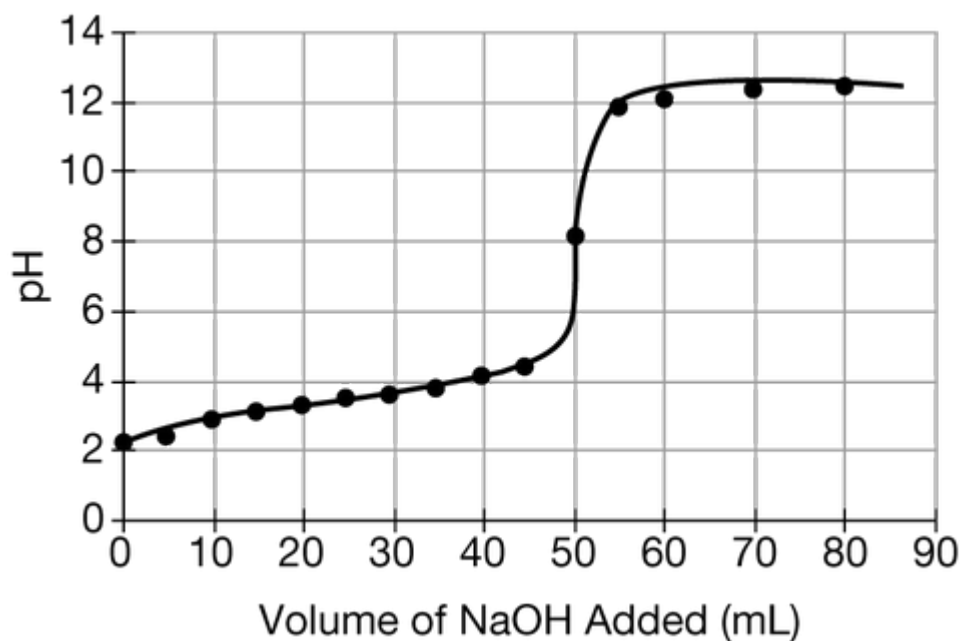
(C) $[\text{ClO}^-] < [\text{HClO}]$

(D) It is not possible to compare the concentrations without knowing the $\text{p}K_b$ of ClO^- .



Unit 8 Progress Check: MCQ

23.



HCNO is a weak acid with a $\text{p}K_a$ value of 3.5. The graph above shows the results of a titration of an aqueous solution of **HCNO** with 0.100 M NaOH . Based on the results, the concentration of CNO^- is greater than the concentration of **HCNO** at which of the following pH values?

(A) 2.0

(B) 3.0

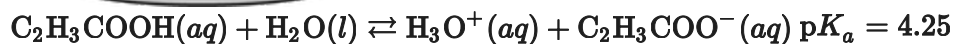
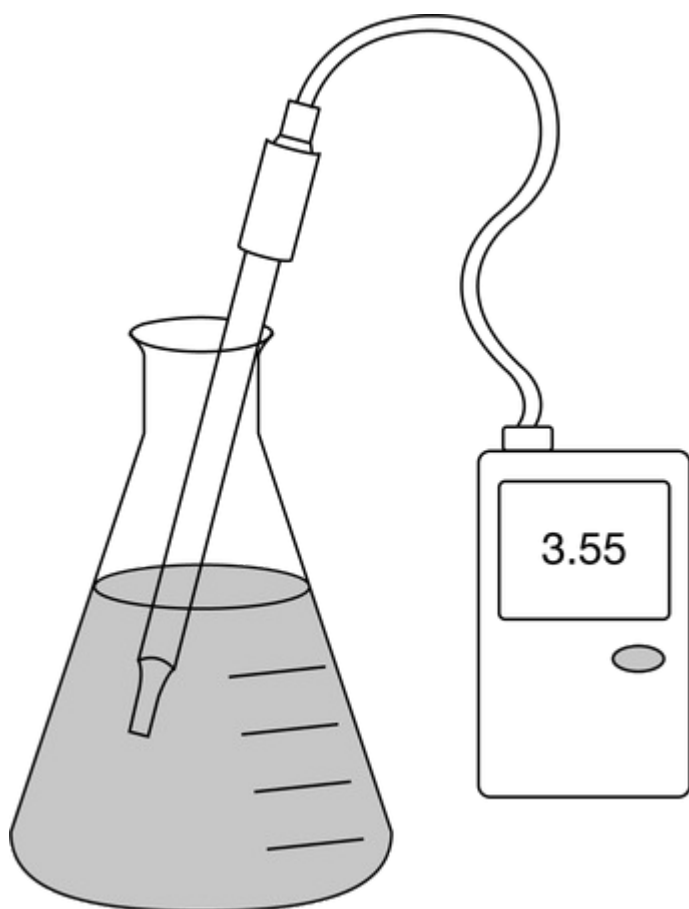
(C) 3.5

(D) 4.0



Unit 8 Progress Check: MCQ

24.



The weak acid ionization equilibrium for $\text{C}_2\text{H}_3\text{COOH}$ is represented by the equation above. A student measures the pH of $\text{C}_2\text{H}_3\text{COOH}(aq)$ using a probe and a pH meter in the experimental setup shown. Based on the information given, which of the following is true?

- (A) $[\text{C}_2\text{H}_3\text{COOH}] > [\text{C}_2\text{H}_3\text{COO}^-]$ since the $\text{p}K_a$ of the weak acid is less than $\text{p}K_w$.
- (B) $[\text{C}_2\text{H}_3\text{COOH}] > [\text{C}_2\text{H}_3\text{COO}^-]$ since the $\text{p}K_a$ of the weak acid is greater than the pH of the solution. ✓
- (C) $[\text{C}_2\text{H}_3\text{COOH}] < [\text{C}_2\text{H}_3\text{COO}^-]$ since the $\text{p}K_b$ for $\text{C}_2\text{H}_3\text{COO}^-$ is less than $\text{p}K_w$.
- (D) $[\text{C}_2\text{H}_3\text{COOH}] < [\text{C}_2\text{H}_3\text{COO}^-]$ since the K_b for $\text{C}_2\text{H}_3\text{COO}^-$ is less than the K_a for $\text{C}_2\text{H}_3\text{COOH}$.



Unit 8 Progress Check: MCQ

25. $\text{CH}_3\text{CH}_2\text{COOH}(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{H}_3\text{O}^+(aq) + \text{CH}_3\text{CH}_2\text{COO}^-(aq)$ $K_a = 1.4 \times 10^{-5}$ at 25°C
 $\text{CH}_3\text{CH}_2\text{COO}^-(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{CH}_3\text{CH}_2\text{COOH}(aq) + \text{OH}^-(aq)$ $K_b = 7.4 \times 10^{-10}$ at 25°C

The acid equilibrium for $\text{CH}_3\text{CH}_2\text{COOH}$ and the base equilibrium for $\text{CH}_3\text{CH}_2\text{COO}^-$ are represented above. One liter of a buffer solution with $\text{pH} = 4.85$ is made by mixing 0.100 M $\text{CH}_3\text{CH}_2\text{COOH}$ and 0.100 M $\text{NaCH}_3\text{CH}_2\text{COO}$. If 10.0 mL of 0.500 M NaOH is added to the buffer, which of the following is most likely the resulting pH , and why?

- (A) The pH will be much less than 4.85, because the buffer will respond to the addition of NaOH by producing more $\text{CH}_3\text{CH}_2\text{COOH}$.
- (B) The pH will be much greater than 4.85, because there is a large increase in the concentration of the weak base, $\text{CH}_3\text{CH}_2\text{COO}^-$.
- (C) The pH will be slightly less than 4.85, because the addition of NaOH reduces the autoionization of H_2O .

- (D) The pH will be slightly greater than 4.85, because some $\text{CH}_3\text{CH}_2\text{COOH}$ will react with the added bases, resulting in a slight decrease in $[\text{H}_3\text{O}^+]$. ✓

26. A buffer solution that is 0.100 M in both HCOOH and HCOO^- has a $\text{pH} = 3.75$. A student says that if a very small amount of 0.100 M HCl is added to the buffer, the pH will decrease by a very small amount. Which of the following best supports the student's claim?

- (A) HCOO^- will accept a proton from HCl to produce more HCOOH and H_2O . ✓

- (B) HCOOH will accept a proton from HCl to produce more HCOO^- and H_2O .

- (C) HCOO^- will donate a proton to HCl to produce more HCOOH and H_2O .

- (D) HCOOH will donate a proton to HCl to produce more HCOO^- and H_2O .

27. $\text{CH}_3\text{COOH}(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{H}_3\text{O}^+(aq) + \text{CH}_3\text{COO}^-(aq)$ $\text{p}K_a = 4.76$ at 25°C

The equilibrium representing the acid dissociation of CH_3COOH is shown above. A buffer solution is prepared by adding 0.10 mol of $\text{NaOH}(s)$ to 1.00 L of 0.30 M CH_3COOH . Assuming the change in volume is negligible, which of the following expressions will give the pH of the resulting buffer at 25°C ?



Unit 8 Progress Check: MCQ

(A) $\text{pH} = 4.76 + \log \frac{0.10}{0.30}$

(B) $\text{pH} = 4.76 + \log \frac{0.10}{0.20}$ ✓

(C) $\text{pH} = 4.76 + \log \frac{0.20}{0.10}$

(D) $\text{pH} = 4.76 + \log \frac{0.30}{0.10}$

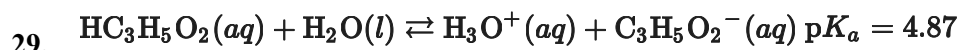
28. To prepare a buffer solution for an experiment, a student measured out **53.49 g** of $\text{NH}_4\text{Cl}(s)$ (molar mass **53.49 g/mol**) and added it to **1.0 L** of **1.0 M** $\text{NH}_3(aq)$. However, in the process of adding the $\text{NH}_4\text{Cl}(s)$ to the $\text{NH}_3(aq)$, the student spilled some of the $\text{NH}_4\text{Cl}(s)$ onto the bench top. As a result, only about **50. g** of $\text{NH}_4\text{Cl}(s)$ was actually added to the **1.0 M** $\text{NH}_3(aq)$. Which of the following best describes how the buffer capacity of the solution is affected as a result of the spill?

(A) The solution has a greater buffer capacity for the addition of base than for acid, because $[\text{NH}_3] < [\text{NH}_4^+]$.

(B) The solution has a greater buffer capacity for the addition of base than for acid, because $[\text{NH}_3] > [\text{NH}_4^+]$.

(C) The solution has a greater buffer capacity for the addition of acid than for base, because $[\text{NH}_3] < [\text{NH}_4^+]$.

(D) The solution has a greater buffer capacity for the addition of acid than for base, because $[\text{NH}_3] > [\text{NH}_4^+]$. ✓



The acid ionization equilibrium for $\text{HC}_3\text{H}_5\text{O}_2$ is represented by the equation above. A mixture of **1.00 L** of **0.100 M** $\text{HC}_3\text{H}_5\text{O}_2$ and **0.500 L** of **0.100 M** NaOH will produce a buffer solution with a **pH = 4.87**. If the NaOH solution was mislabeled and was **1.00 M** instead of **0.100 M**, which of the following would be true?



Unit 8 Progress Check: MCQ

- (A) The **pH** of the resulting solution would still be 4.87 because buffer solutions regulate changes in **pH** regardless of how much **NaOH** is added.
- (B) The **pH** of the resulting solution would be somewhat higher than 4.87 because adding **NaOH** that is 10 times more concentrated makes the concentration of the conjugate base 10 times larger.
- (C) The **pH** of the resulting solution would be much higher than 4.87 because the weak acid would be completely neutralized by the larger amount of **NaOH** added. ✓
- (D) The **pH** of the resulting solution would still be 4.87 because the solution contains a large volume of the weak acid to react with the **NaOH** and maintain the same **pH**.

30. $\text{HNO}_2(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{H}_3\text{O}^+(aq) + \text{NO}_2^-(aq) \quad \text{p}K_a = 3.25$

Sample	Solution Composition	pH
1	25.0 mL buffer solution	3.26
2	25.0 mL buffer solution	3.25
3	25.0 mL buffer solution and one drop of HCl	3.03
4	25.0 mL buffer solution	3.21
5	25.0 mL buffer solution	3.25

The acid ionization equilibrium for **HNO₂** is represented by the equation above. A **250.0 mL** buffer solution is prepared by mixing **125.0 mL** of **0.20 M HNO₂** and **125.0 mL** of **0.1 M NaOH**. To test the buffer capacity, the **pH** is measured and recorded in the table for four samples of the buffer and one sample of a mixture of the buffer and **HCl**. Which of the following best helps explain why the **pH** of sample 4 is lower than the pH of the other samples containing only buffer solution?

- (A) Prior to measuring the **pH** of sample 4, some water evaporated, resulting in an increase in the concentration of **HNO₂** in the sample.
- (B) After measuring the **pH** of the more acidic sample 3, the **pH** probe was not rinsed and wiped, resulting in the neutralization of a very small amount of the conjugate base in sample 4. ✓
- (C) Prior to measuring the **pH** of sample 4, the room temperature dropped suddenly, resulting in an increase in the **pK_a** for **HNO₂**.
- (D) The volume used to measure the **pH** of sample 4 was less than **25.0 mL**, resulting in a decrease in the concentration of **NO₂⁻**.